1 List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.

| IP Class | IP Address Range | Real-World Example |
|---|---|---|
| Class A | 1.0.0.0 – 126.255.255.255 | A large organization's internal network. |
| Class B | 128.0.0.0 – 191.255.255.255 | A university campus network. |
| Class C | 192.0.0.0 – 223.255.255.255 | A small office or home network using a public Class C address block. |
| Class D | 224.0.0.0 – 239.255.255.255 | Multicast video streaming or video conferencing. |
| Class E | 240.0.0.0 – 255.255.255.255 | Reserved for experimental or future use, not normal end-user devices. |

2 Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses.<br><br>Hint: Show your working steps for each calculation.

# Given Information

- IP Address: 192.168.10.45
- Subnet Mask: 255.255.255.0 (/24)

### Step 1: Calculate the Network Address

Network address = IP address AND Subnet Mask

```
IP Address:     192.168.10.45
Subnet Mask:    255.255.255.0
Network ID:     192.168.10.0
```

Network Address: 192.168.10.0

### Step 2: Calculate the Broadcast Address

Since the subnet mask is `/24`, the last 8 bits are for host addresses. Set all host bits to `1`.

```
Network:        192.168.10.0
Broadcast:      192.168.10.255
```

Broadcast Address: 192.168.10.255

### Step 3: Calculate the Valid Host Range

- First valid host = Network address + 1
- Last valid host = Broadcast address − 1

```
First Host: 192.168.10.1
Last Host:  192.168.10.254
```

Valid Host Range: 192.168.10.1 – 192.168.10.254

**Final Answer**

| Item | Answer |
|---|---|
| Network Address | 192.168.10.0 |
| Broadcast Address | 192.168.10.255 |
| First Valid Host | 192.168.10.1 |
| Last Valid Host | 192.168.10.254 |
| Total Valid Hosts | 254 |

Formula: Total valid hosts = $2^h - 2$, where $h$ is the number of host bits.

Here, $2^8 - 2 = 254$ valid host addresses.

3. Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.

## ➢ Subnetting Plan for a Food Delivery Startup Like Zomato

We have the network 10.0.0.0/24 and need 4 separate subnets for Delivery Partners, Restaurants, Customers, and Admins.

To create 4 subnets, we borrow 2 bits from the host portion:

- Original prefix: `/24`
- New prefix: `/26`
- Subnet mask: `255.255.255.192`
- Each subnet has 64 IP addresses, with 62 usable host addresses.

| No. | Group | Network Address | Subnet Mask | Valid Host Range |
|---|---|---|---|---|
| 1 | Delivery Partners | 10.0.0.0/26 | 255.255.255.192 | 10.0.0.1 – 10.0.0.62 |
| 2 | Restaurants | 10.0.0.64/26 | 255.255.255.192 | 10.0.0.65 – 10.0.0.126 |
| 3 | Customers | 10.0.0.128/26 | 255.255.255.192 | 10.0.0.129 – 10.0.0.190 |
| 4 | Admins | 10.0.0.192/26 | 255.255.255.192 | 10.0.0.193 – 10.0.0.254 |

**Calculation :**

Number of subnets:

$$ 2^2 = 4 $$

Usable hosts per subnet:

$$ 2^6 - 2 = 62 $$

Conclusion: The /24 network is divided into four /26 subnets, providing separate IP address ranges for each group. This helps organize network traffic and improve security through appropriate access controls.

4 You are given the following scenario: A college WiFi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet.<br><br>Constraint: Show your calculation steps and final answers.

**Given Information**

- Network: `172.16.0.0/16`
- Required subnets: At least 20

**Step 1: Calculate the bits required for subnets**

Formula:

$$ 2^n \geq \text{Required Subnets} $$
$$ 2^5 = 32 \geq 20 $$

So, we need to borrow 5 bits from the host portion.

**Step 2: Calculate the new subnet mask**

Original prefix: `/16`

Borrowed bits: 5

$$ 16+5=21 $$

New prefix: /21

New subnet mask: 255.255.248.0

**Step 3: Calculate hosts per subnet**

Original host bits = $32-16=16$

Remaining host bits:

\[ 16-5=11 \]

Formula:

\[ 2^h-2 \]
\[ 2^{11}-2=2048-2=2046 \]

Therefore, each subnet has 2,046 usable host addresses.

| Item | Answer |
|---|---|
| Original Network | 172.16.0.0/16 |
| Required Subnets | At least 20 |
| Bits Borrowed | 5 |
| Total Subnets Created | 32 |
| New Subnet Mask | 255.255.248.0 |
| New CIDR Prefix | /21 |
| Usable Hosts per Subnet | 2,046 |

**Conclusion :** The college can create 32 subnets using the /21 subnet mask, with 2,046 usable host addresses in each subnet. This meets the requirement of at least 20 subnets.

5 A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.

- Answer :

The IP address 192.168.1.300 is invalid because each octet in an IPv4 address must be between 0 and 255. Here, `300` exceeds the allowed range.

A valid IP address in the same `/24` subnet would be 192.168.1.100, with subnet mask `255.255.255.0`, provided the address is not already in use.

Conclusion: Use a valid IPv4 address with each octet between 0 and 255.